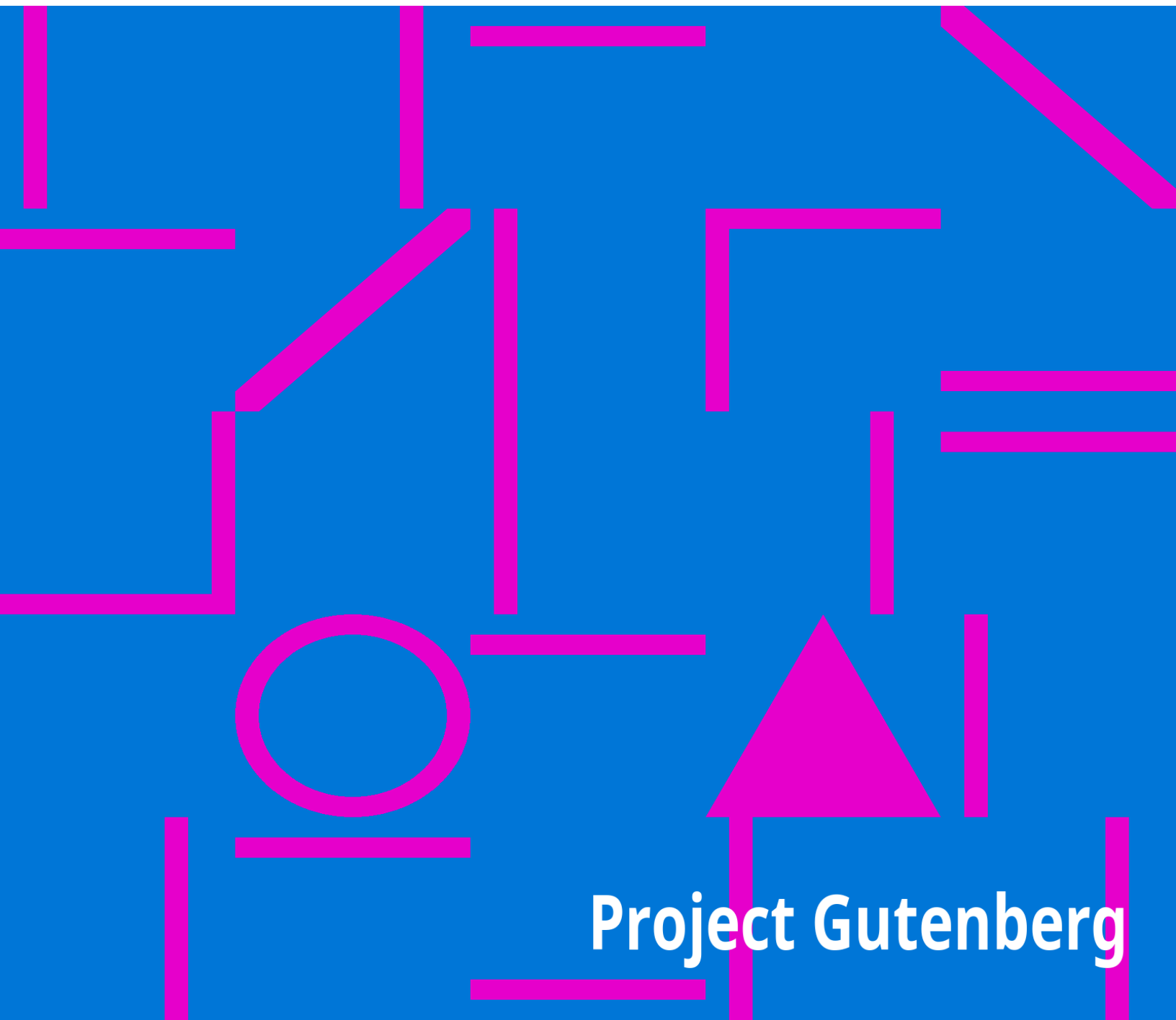


The Mentor: The Weather

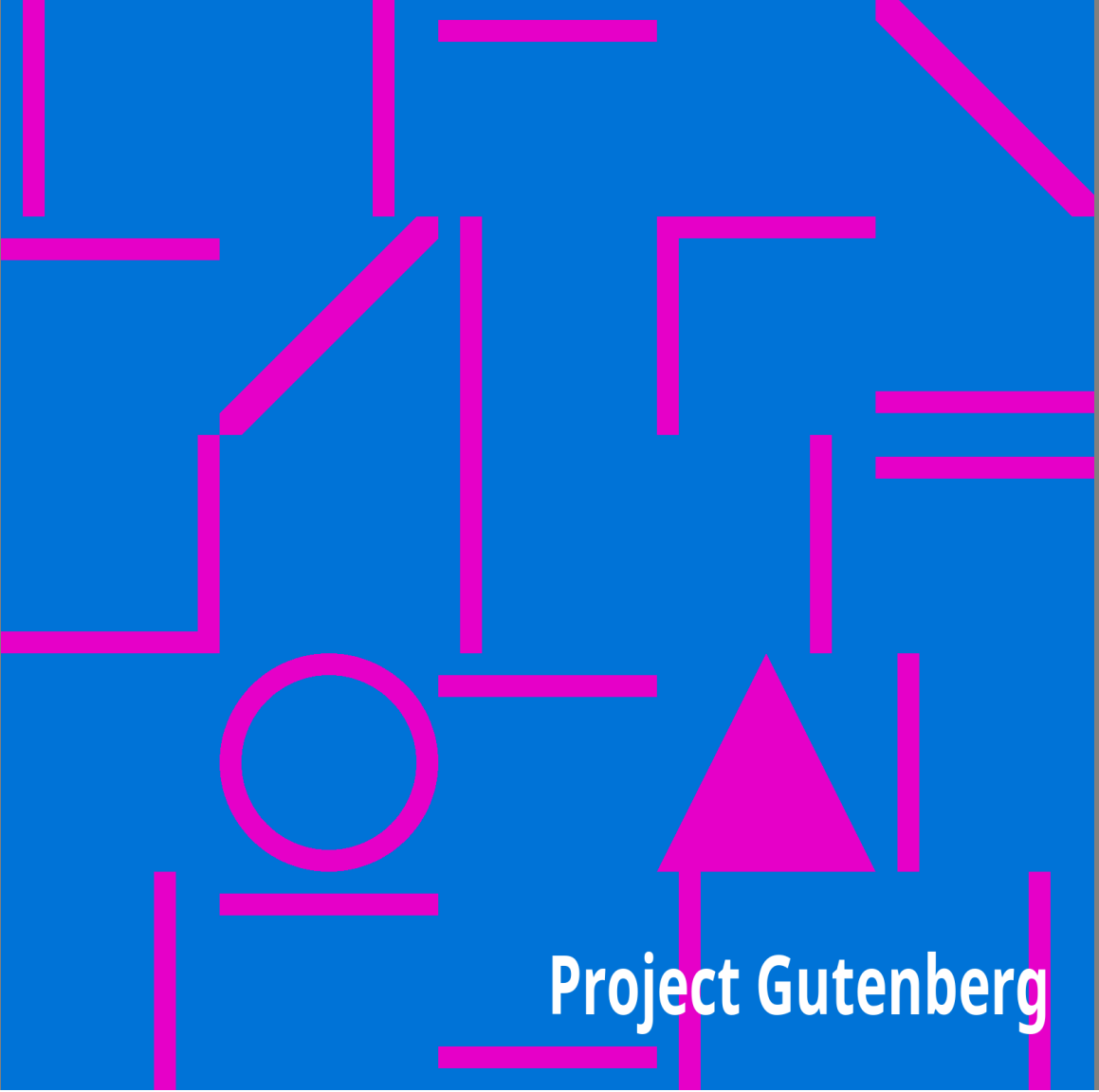
Charles Fitzhugh Talman

The background of the lower half of the image is a solid blue color. Overlaid on this background is a complex, abstract pattern of magenta (bright pink) geometric shapes. These shapes include various line segments of different lengths and orientations, some forming right angles, and a few larger shapes like a circle and a triangle. The pattern is dense and non-repeating, creating a modern, graphic look.

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The Mentor: The Weather

Charles Fitzhugh Talman



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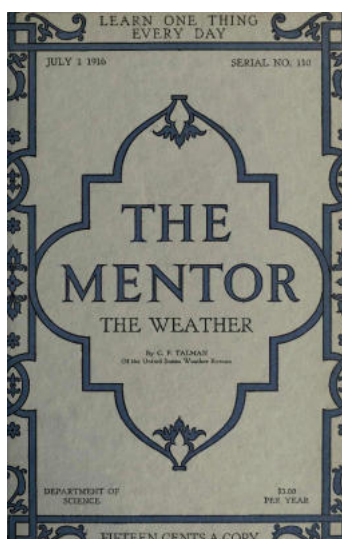
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**THE
MENTOR**

THE WEATHER

By C. F. TALMAN
Of The United States Weather Bureau

DEPARTMENT OF
SCIENCE

\$3.00
PER YEAR

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Old Probabilities



Shall tomorrow's weather be fair or foul? Blow wind—blow moistly from the South, for I go afishing. "Nay, good friend," exclaims the golfer, "the day must be dry and the wind in the west." The farmer moistens his finger and points it toward the sky. "Rain, come, quickly, for my crops," is his prayer. But the maiden's voice is full of pleading: "Let the sun shine tomorrow that my heart may be light on my wedding day."



And so, through the days and seasons, humanity with all its varied needs, turns anxiously, entreatingly to Old Probabilities. And how is it possible for him to satisfy the conflicting demand? He may, on the same day, please the farmer in the West, the fisherman in the South, the golfer in the northern hills, and the bride in the eastern town. But how can he suit them all in one locality on a single day? Old Probabilities is willing and he loves humanity, but his powers and privileges are limited. There are those who say that it is due to the kind endeavors of Old Probabilities to satisfy everybody that our weather has at times become so strangely mixed.



Old Probabilities is a gentle family name and came out of the affection of the people. The name was a matter of pleasantry. It was given to the Chief of the United States Weather Bureau when the department was first established by Congress, and its source lay in the phrase, "It is probable," with which all the weather predictions began. But Old Probabilities, genial prophet and lover of his fellow men, is passing away, for the officer who organized the Weather Bureau became in time displeased with the name and changed the form of the daily prediction so as to read, "The indications are." The phrase is formal and severe. There is naught but cold comfort in it. Our hearts turn back fondly to Old Probabilities and his friendly assurance: "It is *probable* that tomorrow will be fair."



Chickamauga Park, Tenn., in an Ice Storm

THE WEATHER

By CHARLES FITZHUGH TALMAN

Librarian of the U. S. Weather Bureau



THE MENTOR · DEPARTMENT OF SCIENCE · JULY 1, 1916

MENTOR GRAVURES

CENTRAL OFFICE OF THE U. S. WEATHER BUREAU,
WASHINGTON, D. C.
A SIMPLE WEATHER STATION
A MAJESTIC CUMULUS CLOUD
THE OBSERVATORY ON MONTE ROSA
LAUNCHING A METEOROLOGICAL KITE
THE EFFECTS OF SNOW AND ICE—THE CAMPUS, PRINCETON
UNIVERSITY



It is easy to lay too much stress upon the unimportant aspects of weather. It furnishes a bit of conversation over the teacups; it accentuates the twinges of rheumatism; it spoils a holiday. All this, however, is mere byplay.

The real work of the weather—the work that explains the existence of costly weather bureaus, such as the one upon which our Government spends more than a million and a half dollars annually—is momentous beyond calculation. Consider such facts and figures as these:



The head of the British Meteorological Office recently declared that bad weather costs the farmers of the British Isles about one hundred million dollars a year. In our own country it has been estimated that a difference of one inch in the rainfall occurring during July in six States means a difference of two hundred and fifty million dollars in the value of the corn (maize) crop. The world over, the damage wrought by hail-storms is said to average about two hundred million dollars a year. In the city of Galveston a single hurricane once destroyed twenty million dollars' worth of property and six thousand human lives. Thus we might proceed indefinitely.

The fact is that man's welfare is conditioned to an enormous extent and in an endless variety of ways by the vicissitudes of the atmosphere; hence the study of weather—meteorology—is one of the most important of sciences. It is also one of the most strikingly neglected!

At the office of the Weather Bureau in Washington there is a meteorological library of some thirty-five thousand volumes. But meteorological libraries are rare; meteorological books are scarce in other libraries; and meteorologists are so uncommon that whoever declares himself one is likely to be asked, "What is a meteorologist?"

The "meteors" studied by the meteorologist are not shooting stars, but the phenomena of the atmosphere,—rain and snow, cloud and fog, wind and sunshine, and whatever else enters into the composition of weather and climate.

STATIONS OF THE UNITED STATES WEATHER BUREAU

Showing two extreme types: one, an office on the twenty-ninth floor of the Whitehall Building, New York City, with instruments installed on the roof; the other, an independent observatory building, with free exposure on all sides, at St. Joseph, Mo.

THE ATMOSPHERE

The ocean of air in which human beings live, even as deep-sea fishes live at the bottom of the liquid ocean, is called the *atmosphere*. Unlike the liquid ocean, it diminishes rapidly in density from the bottom upward. At an altitude of three and one-half miles it is only half as dense as at sea-level. This is higher than the highest permanent habitations of man. Mountain-climbers and balloonists have attained greater altitudes; but above a level of about five miles the air is too greatly rarefied to support life. Balloonists who ascend still higher must carry a supply of oxygen with them. A little above the ten-mile level the air is only one-eighth as dense as at sea-level. The atmosphere extends at least 300 miles above the earth, at which height its density is computed to be only one two-millionth as great as at sea-level.

The weather with which human beings are concerned may be said to extend upward seven or eight miles; *i.e.*, to the level of the higher clouds. The layer of the atmosphere lying between sea-level and the upper cloud level has certain characteristics that distinguish it from the air above it, and is known as the *troposphere*.

The heating of the atmosphere by the sun is the beginning of all weather, and the temperature of the air is the most important weather element. As soon as we begin to study atmospheric temperature, we encounter a paradox. The heat of the air is all derived from the sun (except a minute quantity from the interior of the earth, and an infinitesimal quantity from other heavenly bodies), and it would therefore seem at first glance that the upper layers of the atmosphere should be warmer than the lower. Experience proves the reverse to be the case. A mountain overgrown with tropical vegetation on its lower slopes is, if high enough, crowned with eternal snows. A thermometer carried upward in the air shows under average conditions a fall of temperature of one degree (Fahrenheit) for every 300 feet of ascent. This fall of temperature with ascent continues to the upper limit of the troposphere, where the average temperature is something like 70 degrees below zero.

THE NEW IDEA IN WEATHER OBSERVATORIES

The Observatory of the Ebro (Spain), founded by Spanish Jesuits, is devoted to studying the interrelations of sun, earth and air. Its admirable equipment includes apparatus for the direct and spectroscopic study of the sun, for measuring solar radiation, atmospheric electricity, earth currents, terrestrial magnetism, and earthquakes; besides the ordinary routine of a meteorological observatory. The results of all these observations are published side by side, to facilitate comparison.



Above the troposphere is a region called the *stratosphere*, or *isothermal layer*, in which an ascending thermometer shows irregular and generally small changes of temperature—not infrequently a rise of temperature with ascent. The exploration of the stratosphere is one of the most fascinating fields of meteorological research, but lies somewhat beyond the scope of an essay on weather. It is carried out chiefly with the aid of small free balloons, some of which (sounding balloons) bear self-registering thermometers and other instruments, while others (pilot balloons) bear no instruments, but show by their movements the drift of the air currents. The greatest altitude ever attained by a sounding-balloon was 21.8 miles; by a pilot-balloon, 24.2 miles. The branch of meteorology dealing with the study of the upper air is called *aërology*.



**A LONELY OUTPOST ON THE VERGE OF
THE ANTARCTIC**

The Argentine meteorological station in the South Orkneys. Once a year an expedition is sent from Buenos Aires to relieve the staff of four observers. This is the southernmost permanently inhabited spot

on the globe; and it has not even wireless communication with the rest of the world.

Reverting to the temperature of man's environment, the reason why the atmosphere is warmest at the bottom is this: The sun's rays come to us from outer space in the form of vibrations in the ether, and warm the air to only a slight extent in passing through it. They are absorbed by the ground, and converted into heat waves. The air is then warmed by contact with the warm ground. Lastly, the warming of the lower air gives rise to air-currents, which distribute the heat through the atmosphere.

BAROMETRIC PRESSURE

If our weather were uniform, it would furnish little matter for conversation; in fact, would hardly be weather at all. Changeableness is the salient feature of weather, and to understand weather changes one must know something about barometric pressure.

Like all other forms of matter, the invisible air has weight. At sea-level it exerts a downward pressure averaging 14.7 pounds to the square inch. Atmospheric pressure is measured by means of an instrument called the *barometer*, in which the weight of the air is balanced against a column of mercury. As the height of the mercurial column varies with the pressure of the air, and is taken as the measure of the latter, we follow the practice of expressing pressure (a force) in linear units (inches or millimeters). This practice is retained even in the use of the aneroid barometer, which contains no mercurial column. Hence, when we say that the average barometric pressure at sea-level is 29.92 "inches," we are really expressing in a roundabout way the weight of the air at that level.

Courtesy of U. S. Bureau of Standards and Popular Science Monthly.



a



b

HOW THE CAMERA ANALYZES LIGHTNING

The same flashes photographed with (a) a stationary camera, and (b) a camera revolving on a vertical axis. One of the flashes is seen to have consisted of several successive discharges along an identical path

Barometric pressure not only varies somewhat regularly with altitude—diminishing as we ascend—but also less regularly from place to place in a horizontal direction, and from time to time at a given place. In studying the weather meteorologists frequently wish to compare the barometric pressures prevailing at a certain time at a number of places lying in the same horizontal plane. Given a system of meteorological stations scattered over a certain territory, the first step is to secure simultaneous readings of the barometers at these stations. Then, if the stations are at various altitudes, as they commonly are, corrections must be applied to the readings to reduce all to a common plane; the plane adopted for this purpose is sea-level. Since most stations are *above* sea-level, and since atmospheric pressure diminishes with altitude, reduction to sea-level generally involves applying an *additive* correction.

THE WEATHER MAP



**ASCENT OF A SOUNDING
BALLOON**

The first made in the United States;
at St. Louis, Mo., in 1904

Now please attend carefully to what follows; because I am going to attempt to put into a minimum number of words the essential facts concerning the *weather map*, the best clue to weather mysteries yet devised by man.

At about 200 stations of the Weather Bureau, distributed over the United States, the barometer and other meteorological instruments are read twice a day; viz., at 8 A. M. and 8 P. M., eastern standard time. The readings are promptly telegraphed in cipher to Washington, where they are entered on a map.

The barometer readings at the different stations, reduced to sea-level as just explained, will vary, say, from 29 to 31 inches. Lines, called *isobars*, are now drawn through places having the same pressure; the intervals between the lines corresponding to differences in pressure of one-tenth of an inch. Lines (*isotherms*) are also drawn to connect places having the same temperature, a little arrow at each station shows the direction of the wind at that point, and various other symbols are used to facilitate the interpretation of the map; but the isobars are more important than anything else.

Here is the weather map for the morning of January 9, 1886. The solid curved lines are isobars, representing barometric pressures ranging all the way from 28.7 to 30.8 inches. It will be seen at a glance that these lines tend to assume roughly circular forms, inclosing regions where the pressure is lower or higher than the average. Moreover, the little arrows (which "fly with the wind") show that the winds round a center of low pressure tend to blow in a direction contrary to that followed by the hands of a clock (in the southern hemisphere the reverse is true), but instead of blowing in circles are inclined somewhat inward toward the center. Round a center of high pressure (in the northern hemisphere) the typical circulation of the winds is exactly opposite ("clockwise," and inclined outward), though the accompanying map does not show this particularly well.



CIRRO-STRATUS

The appearance of this cloud precedes by a day or so the arrival of rainy and stormy weather

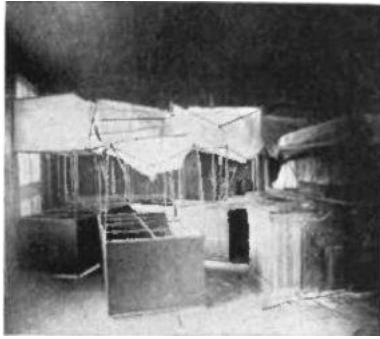


ALTO-CUMULUS



FAIR WEATHER CUMULUS

This cloud marks the summit of an ascending air current, and appears toward midday or early afternoon in the warm season. When the air rises powerfully to great heights, cumulus is built up in mountainous masses and may become cumulo-nimbus, the thundercloud.

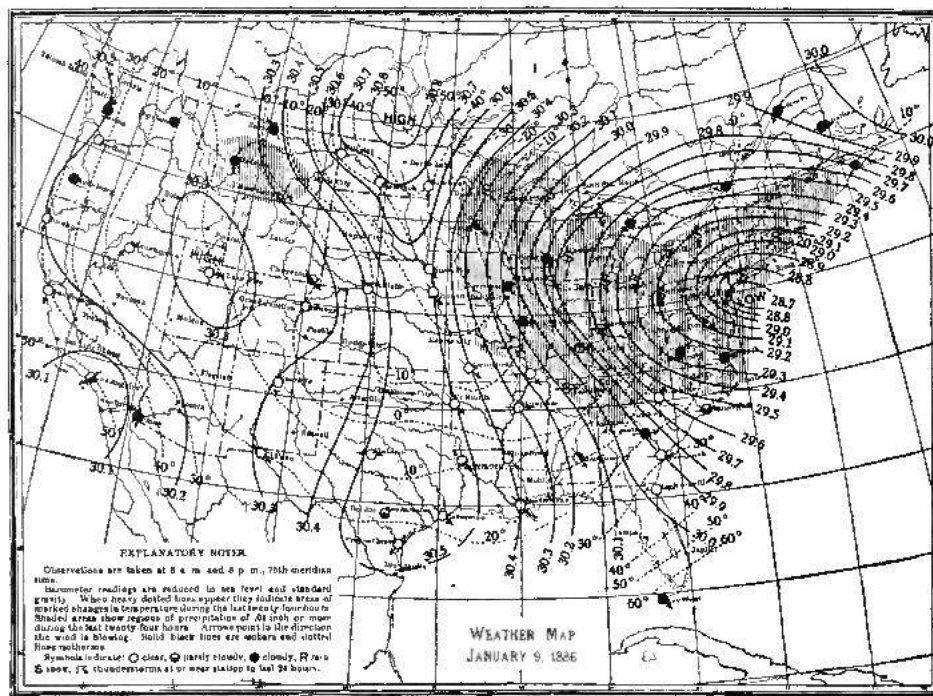


**THE KITE HOUSE AT AN
AEROLOGICAL OBSERVATORY**

Some of the kites are much the worse for wear
after flying in a storm

An area of low pressure, with its system of winds, is called a *cyclone*, or *low*. An area of high pressure, with its system of winds, is called an *anticyclone*, or *high*. Note that a cyclone is not necessarily a storm, though the one shown on this map, with its center not far from New York City, was a very violent storm, which, when this map was drawn, was sweeping up the Atlantic coast. (Popular usage applies the term "cyclone" to the tornado.) The strength of the winds in a cyclone depends upon the contrast in barometric pressure between its center and its outer border. A cyclone with crowded isobars always has strong winds; when the isobars are widely spaced the winds are gentle.

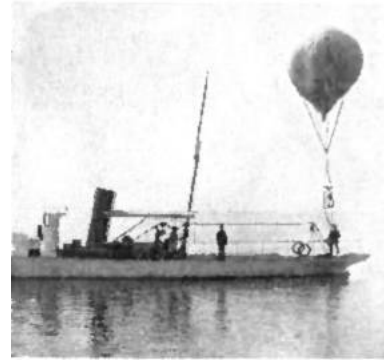
These areas of low and high pressure, in addition to their movements about their centers, move bodily across the country, in a general west-to-east direction, at an average speed of over 500 miles a day. This double movement may be compared to that of a carriage-wheel, rotating and advancing at the same time. Most of our cyclones enter the country from the Canadian North-west—though many come from other regions—and nearly all of them pass off to sea in the neighborhood of the Gulf of St. Lawrence. Their route across the country varies greatly, depending in part upon the season.



**WEATHER MAP
JANUARY 9, 1886**

THE WEATHER IN CYCLONES AND ANTICYCLONES

Barometric pressure is not an element of weather, in the ordinary sense of the term, since the fluctuations of pressure that occur in the human environment are entirely inappreciable to the senses. We have seen, however, that pressure is intimately related to wind, which is a weather element of much importance. In noting that systems of high and low pressure are constantly traveling across the country, and that they are accompanied by winds having fairly definite characteristics in relation to each, we have taken an important step toward bringing order out of the (to the uninitiated) chaotic sequence of weather. Obviously, a system of telegraphic weather reports makes it possible to keep close watch of these wind systems, and, from their locations on today's weather map, to form some idea where they will be tomorrow. Thus the weather forecaster is enabled to give notice of the imminence of those violent winds that destroy life and property at sea, and, to a less extent, on land. There is an element of uncertainty in such predictions—since storms, unlike railway trains, are not confined to fixed routes and regular schedules—but the practised forecaster acquires an instinct that helps him to forestall their vagaries.



**SENDING UP A METEOROLOGICAL
BALLOON ON LAKE CONSTANCE**

Between Switzerland and Germany.



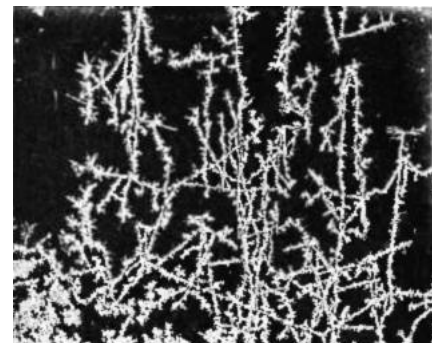
**MARVIN RAIN AND SNOW
GAGE**

With trumpet-shaped wind-shield at top. In the middle is seen the cylindrical collector. This is removed and weighed with its contents to ascertain the amount of rain or snow that has fallen

Now what is true of wind is also true to a certain extent of the other elements of weather,—they bear typical relations to the distribution of atmospheric pressure. Cyclones are usually preceded by rising temperature and accompanied by cloudiness and rain or snow; anticyclones are usually preceded by falling temperature and attended by fair weather.

Referring again to the map of January 9, 1886, and following the course of the isotherms, or temperature lines, we see that abnormally cold weather prevailed over the Middle Western and Southern States. The isotherm of zero dips far south across northern Texas, Arkansas, Mississippi, Alabama, and Tennessee; while in the upper Mississippi and Missouri Valleys the temperatures were from 20 to 40 degrees below zero. These regions were, in fact, in the grip of a severe "cold wave," which had entered the country a day or two before, preceding the anticyclone here seen central north of Dakota. Cold northwesterly winds were sweeping over the Great Plains, and as far south as the Gulf.

The same map shows typical weather accompanying the cyclone central on the Atlantic coast. From the seaboard west to the Mississippi Valley rain or snow had fallen within the previous twenty-four hours (indicated by shading), and snow (indicated by S) was falling at the moment of observation at a majority of stations within this area. Elsewhere in the same region the weather was cloudy.



HOARFROST

The foregoing remarks indicate in a general way the significance of the weather map and the principles upon which scientific weather

predictions are based. The endless procession of highs and lows brings to any place on the map constant alternations of heat and cold, storm and sunshine. The forecaster watches the procession, and draws his inferences as to what will happen in this or that part of the country within the next day or two (forty-eight hours is about the limit of his outlook).

Minute crystals of ice deposited from the air. Under a magnifying-glass they show a variety of beautiful forms

"Long-range" forecasting is still a thing of the remote future. Forecasts for a week in advance, are, indeed made by the Weather Bureau with the aid of reports from a chain of stations extending round the globe, but these are in very general terms.

In January, 1914, the Bureau began publishing a "daily weather map of the Northern Hemisphere." This publication is, at present, suspended on account of the war.

SOME WEATHER MISCELLANIES



THE EFFECTS OF AN ICE STORM AT CANTON, N. Y.
March 25-27, 1913

It would require a book, rather than a brief essay, to describe all the vicissitudes of weather, and many books that attempt to do this have been written.^[A] We have space here only to mention a few important features of the weather met with in our own country.

The southern and southeastern part of a cyclone, some hundreds of miles from the center, is a favorite breeding-ground for *thunderstorms* and *tornadoes*. Thunderstorms of the type known as "heat thunderstorms" also occur with no special relation to cyclonic centers in regions where the ground has been intensely heated. In either case the storm is built up by rapidly ascending air, which cools and condenses its water vapor, first into enormous clouds (*cumulo-nimbus*, or "*thunderheads*"), and then into rain, frequently

accompanied by hail. It would be necessary to go to some length to explain the familiar electrical manifestations of the thunderstorm—some points, indeed, are not perfectly clear to meteorologists—but it should be stated that these are always the result, not the cause, of the storm. *Lightning* is an electrical discharge between cloud and earth, or cloud and cloud, and *thunder* is simply the violent soundwave set up by the sudden expansion of the heated air along the path of the discharge,—the same acoustic phenomenon that accompanies an ordinary explosion.



SUMMIT HOTEL AT SUMMIT, CAL.

On March 18, 1911. A three-story building whose first story is buried under twenty-six feet of snow

A *tornado* (popularly miscalled a "cyclone") is an extremely violent vortex in the air, usually less than 1,000 feet in diameter.

Courtesy of the Scientific American



A WATERSPOUT NEAR BEAUFORT, N. C., IN
AUGUST, 1911

Besides its very rapid rotary motion, it has a progressive motion at a speed averaging forty or fifty miles an hour. Its position at any moment is marked by a black funnel-shaped cloud, which grows downward

from the sky and does not at all times reach the earth. A waterspout at sea is an identical phenomenon, though usually less violent. Along its narrow path the tornado demolishes everything,—wooden houses are blown to splinters, trees uprooted or stripped of their branches, structures of heavy masonry laid in ruins. Something like a hundred lives are lost each year in these storms, on an average, and one of them (St. Louis, May 27, 1896) destroyed thirteen million dollars' worth of property.

A *blizzard* is a high, cold wind, accompanied by blinding snow, which in winter sometimes blows out of the front of an advancing anticyclone, especially in our North-Central States. A similar wind, with or without snow, is

called in Texas a *norther*.

A *chinook* is a warm, dry wind that descends the eastern slope of the Rocky Mountains in Montana, Wyoming and Colorado, and flows north-eastward over the plains. Its effects are most pronounced in winter, when it brings about a very sudden rise in the temperature—in extreme cases as much as forty degrees in fifteen minutes! It causes snow to vanish as if by magic, and is appropriately nicknamed the "snow-eater."



IN THE WAKE OF A TORNADO

The tornado destroyed a house and barn, but left a path in the center with practically no harm done

up along the Atlantic Coast or crossing the Gulf of Mexico into the southern United States. Soon after entering the temperate zone they increase in size and diminish in violence, but are still vigorous enough on reaching the Gulf or South Atlantic Coast to cause great devastation. Low-lying shores are often inundated by the immense waves they generate.



**LOOKING DOWN ON A SEA OF FOG FROM
MT. TAMALPAIS, CALIFORNIA**

Bureau, which gives prompt notice to the orchardist when such precautions are in order. The bureau publishes charts showing the average and extreme dates of the last frost in spring and the first frost in autumn for all parts of the country.

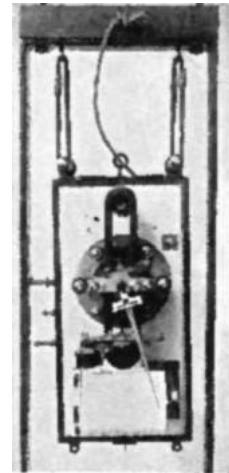
A *fog* is a cloud resting on the surface of the earth. In the United States fog is commonest along the northern and middle parts of the Atlantic and Pacific Coasts. In the interior of the country, especially the western part, it is of rare occurrence, the average number of days a year with fog being less than ten.

"*Cloudburst*" is merely a picturesque name for a very heavy shower; usually a thunder-shower.

West India hurricanes occasionally visit the United States, especially in the late summer and early autumn. These storms begin as violent cyclones of small extent (300 to 600 miles in diameter), usually somewhere east of the West Indies, sweep in a long curve across the Caribbean Sea, and then turn north, either passing

Cold waves are the rapid and severe falls in temperature that sometimes occur in winter, especially at the front of an anticyclone. Warnings of these occurrences, issued by the Weather Bureau twenty-four to thirty-six hours in advance, often result in the saving of millions of dollars' worth of merchandise susceptible to damage by freezing.

Frosts in the spring and autumn are also predicted with great success, to the immense advantage of farmers, market-gardeners, and horticulturists. The practice of smudging or heating orchards, now so widespread, is usually carried on under the advice of the Weather



**TURPAIN'S
THUNDERSTORM
RECORDER**

Or ceraunograph. This is one of several instruments designed to register the natural electric waves, or "strays," which sometimes interfere seriously with the transmission of wireless telegrams. Strays are often generated by lightning discharges, near or distant, and this instrument therefore serves to give notice of an approaching thunderstorm

Lastly—weather fallacies are rife. *Indian summer* is merely a type of mild, hazy, heavenly weather that prevails intermittently during our long American autumns. The *equinoctial storm* is a myth; the climate has not "changed" anywhere within the span of a human lifetime (one year differs from another, but there is no progressive or permanent change); and the *moon* has nothing whatever to do with THE WEATHER.

[\[A\]](#) See "Brief List of Meteorological Textbooks and Reference Books," 3d ed., by C. Fitzhugh Talman. For sale by the Superintendent of Documents, Washington, D. C. Price 5 cents.

SUPPLEMENTARY READING

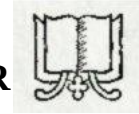
CLIMATE AND WEATHER	<i>By H. N. Dickson</i>
AMERICAN WEATHER	<i>By A. W. Greely</i>
WEATHER SCIENCE	<i>By R. G. K. Lempfert</i>
SOME FACTS ABOUT THE WEATHER	
Second edition.	<i>By W. Marriott</i>
METEOROLOGY	<i>By W. I. Milham</i>
The latest general textbook on the subject in English.	
FORECASTING WEATHER	<i>By W. N. Shaw</i>
ELEMENTARY METEOROLOGY	<i>By F. Waldo</i>

Consult also the numerous publications of the United States Weather Bureau, which will be found in most public libraries.

✱ Information concerning the above books and articles may be had on application to the Editor of The Mentor.



THE OPEN LETTER



"What is lightning and what causes it?" The question came to us a few days after we had made announcement of a "Weather" number of The Mentor. It was a natural question, for lightning is the most sensational of all weather phenomena. It has always had a fearful sort of fascination for humanity. To the ancients it came as a bolt of wrath from the hand of Jove. To the fire-worshippers it was a warning message. To parched travelers it was a bright promise, for it heralded the coming of rain. To the superstitious it was a signal flash from the spirit world. And to those of nervous temperament it was a highly disturbing phenomenon producing emotions varying from uneasiness and alarm to hysteria. The question then, "What is lightning and what causes it?" has an interest for all. I referred it to Mr. Talman, the author of the Mentor article on "The Weather." His reply follows.

"Not so many generations ago 'natural philosophers' thought that inflammable gases, exhaled from the earth, took fire spontaneously in the air, and that this was lightning. The idea also prevailed—and it is not yet quite extinct—that a stroke of lightning involved the hurling down from the sky of a mass of rock, called a 'thunderbolt.' In the eighteenth century people became quite familiar with the process of generating, by friction, a mysterious something called 'electricity,' which, when it passed from one body to another through a small layer of intervening air, produced sparks. Several philosophers noticed the resemblance between these sparks and lightning. It remained, however, for Benjamin Franklin to prove that lightning was really an electrical discharge on a large scale. The experiments by which he proposed to demonstrate this were successfully performed, first by others, in France, and then, by Franklin himself, at Philadelphia. With the aid of his famous kite he drew down from a thundercloud a little of the 'electrical fluid' (as it was then called), and produced tiny sparks from an iron key at the lower end of the wet kite-string.

"We do not even yet know what electricity is, but we know a great deal about the way it behaves and the effects it produces. There are two kinds of electricity, which we call *positive* and *negative*. A body is said to be *charged* when it has an excess of either kind, and the two kinds have a tendency to unite and neutralize each other's effects. Thunderclouds become heavily charged with electricity. We are not quite sure how this happens, but it is now commonly believed that the strong uprising currents of air that occur in the storm, in the process of breaking up the water-drops in the cloud also separate positive from negative electricity; leaving the former in excess in the part of the cloud next to the earth, and carrying the latter far aloft.

"By a process called 'induction' the positive charge in the cloud draws an excess of negative electricity to the surface of the ground underneath. The stronger the contrast between these opposite charges, the harder they try to break through the interposing barrier of the air (which is a poor conductor of electricity) and to neutralize each other. At length they succeed in doing so. A powerful stream of electricity flows for an instant between cloud and earth. Its passage heats the air and makes it luminous—just as the passage of an electric current heats the filament of an electric lamp and makes it luminous. This is lightning.

"These discharges occur not only between the clouds and the earth, but also, and probably more often, between clouds charged with opposite kinds of electricity.

"The sudden expansion of the heated air along the path of the discharge affects our ears just as does the sudden expansion of the air at the mouth of a gun when it is fired. In each case a wave is sent through the air in all directions from the place of disturbance, and our ear-drums are set in vibration. That is thunder."

Take courage then, you timid ones, who wince in the lightning's flash and tremble under the thunder's roll. Thunder is simply a vibration of your ear drums—and, when you hear the thunder, be assured, all danger is over.

A. D. Moffat

EDITOR

THE MENTOR ASSOCIATION

ESTABLISHED FOR THE DEVELOPMENT OF A POPULAR INTEREST
IN ART, LITERATURE, SCIENCE, HISTORY, NATURE, AND TRAVEL



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THE MENTOR

The Mentor Index

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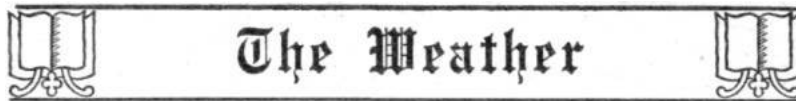
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 - August 1. ARGENTINA. *By E. M. Newman, Lecturer and Traveler.*



CENTRAL OFFICE OF THE UNITED STATES WEATHER BUREAU, WASHINGTON, D. C.



WEATHER SERVICES AT HOME AND ABROAD

Monograph Number One in The Mentor Reading Course



POSTED up in public offices, in hotel corridors, and other conspicuous places in our cities, the official weather map is a familiar sight. Even more familiar is the official weather forecast, displayed, as a rule, on the first page of the daily newspaper, and sent broadcast over the country on the little brown cards which one may see in the village postoffice as well as in the city drug-store. When a great storm sweeps over land or sea, detailed official reports concerning its progress and characteristics are published in the daily press. When a lawsuit involves a dispute as to the temperature or the state of the sky on a certain day, the official weather records are consulted.

How much do you know about the branch of the national government that is charged with the duty of keeping watch of the weather—recording its vagaries as they occur, and also predicting them, as far as is humanly possible?

Besides its office in Washington, where more than two hundred persons are constantly employed, the Weather Bureau has about two hundred stations, manned by professional meteorologists and observers. One of these will be found in almost every large city, while some are in towns of very modest importance. A regular Weather Bureau station is well worth a visit. The instrumental equipment of these stations is almost superhuman in the accuracy with which it sets down on paper the chronicle of weather happenings from day to day and from moment to moment. Little less marvelous is the system by means of which weather information—past, present and future—is disseminated from these official foci. The postoffice, the telephone, the telegraph (wire and wireless) are all pressed into service to the fullest extent—especially in giving timely notice of approaching storms and other destructive forms of weather. These agencies are supplemented by visible and audible signals, in the shape of flags, lanterns, railway whistles and so forth.

Contrary to popular belief, the Weather Bureau does not exist primarily for the purpose of telling the public (with a considerable margin of uncertainty) whether it will be advisable, on the morrow, to carry an umbrella or wear an overcoat. The important work of the Bureau is twofold. It consists, first, in the prediction of those atmospheric visitations, such as storms, floods, and cold waves, which endanger life and property on a large scale; and, second, in the maintenance of the records that form the basis of climatic statistics. In both these directions the Bureau splendidly justifies its existence.

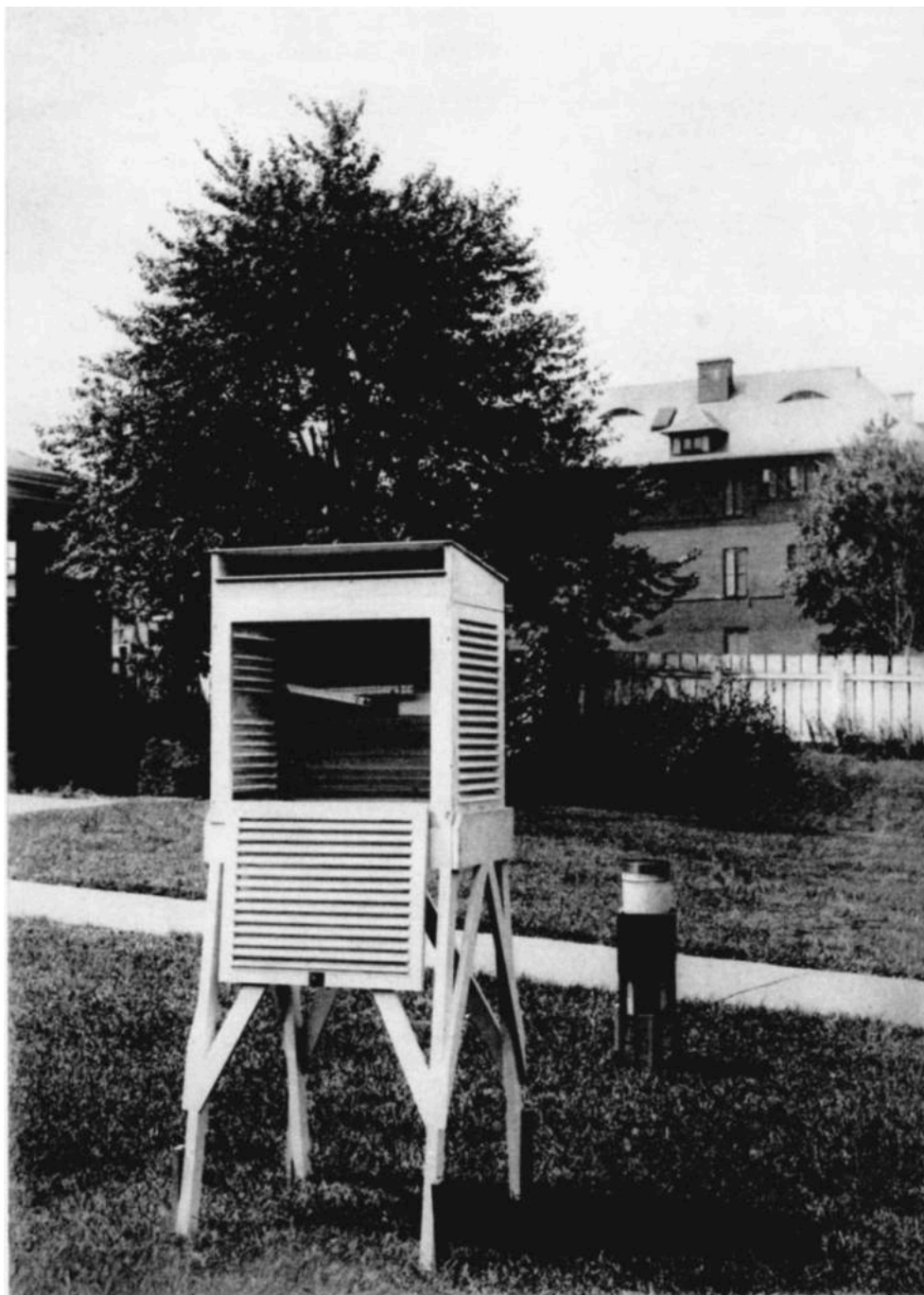
Our national weather service was founded in 1870, and for twenty years was maintained by the Signal Corps of the Army. In 1890 it was established on the present basis, as the Weather Bureau of the Department of Agriculture.

Most civilized countries possess official services for the observation and prediction of weather, though no other is organized on quite so grandiose a scale as ours. The British Meteorological Office, the Prussian Meteorological Institute, the Central Meteorological Bureau of France, and the Central Physical Observatory of Petrograd are among the leading institutions of this character in the Old World. Admirable weather services also exist in India, Japan, Australia, Canada, Argentina and elsewhere.

PREPARED BY THE EDITORIAL STAFF OF THE MENTOR ASSOCIATION

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A SIMPLE WEATHER STATION



The Weather



METEOROLOGICAL INSTRUMENTS

Monograph Number Two in The Mentor Reading Course



HE history of meteorological instruments dates back at least as far as the fourth century before the Christian era, when the depth of rainfall was measured in India by some form of gauge. We again hear of rain-gauges being used in Palestine in the first century of the present era. Thermometers with fixed scales were used in Italy in the seventeenth century, and the great Galileo, born in Pisa in 1564, took part in perfecting these instruments. Wind-vanes were known to the ancients. The earliest one of which we have any record surmounted the famous Tower of the Winds at Athens. In the Middle Ages the weathercock became the usual adornment of church steeples. The barometer was invented by Torricelli in 1643.

Most meteorological instruments, however, are of quite recent origin, and this is true especially of these types of apparatus that make automatic records, thus replacing, to a large extent, the human observer.

Our picture on the other side of this sheet shows the instruments used by the "co-operative" observers of the Weather Bureau. These observers, of whom there are about 4,500, well distributed over the country, serve the government without pay, and their painstaking observations have alone made possible a detailed survey of our climate. In the picture we see, on the right, an ordinary rain-gauge, and, on the left, a thermometer-screen containing two thermometers; viz., a maximum thermometer, for recording the highest temperature of the day, and a minimum thermometer, for recording the lowest. The screen, which is of wood, painted white, serves to shield the instruments from the rays of the sun, while permitting free ventilation. Under these conditions the thermometers show the temperature of the *air*; whereas when exposed to direct sunlight a thermometer shows the temperature acquired by the instrument itself, and this may differ materially from the air temperature.

In contrast to this simple equipment, we find at a regular meteorological station, or observatory, an impressive collection of apparatus for observing and recording nearly all the elements of weather. The pressure of the air is measured by the mercurial barometer, and registered continuously by the barograph; the temperature of the air is automatically recorded by the thermograph. Other self-registering instruments maintain continuous records of the force and direction of the wind, the amount and duration of rainfall, the duration of sunshine, the humidity of the air, etc. There are also instruments for measuring evaporation, the height and movement of clouds, the intensity of solar radiation, the elements of atmospheric electricity, and various other phenomena of the atmosphere.

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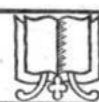
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A MAJESTIC CUMULUS CLOUD



The Weather



CLOUDS AND RAINFALL

Monograph Number Three in The Mentor Reading Course



HE International Cloud Classification, now generally used by meteorologists, is an amplification of one introduced by an ingenious English Quaker, Luke Howard, in the year 1803. Howard distinguished seven types of cloud, to which he gave the Latin names *cirrus*, *cumulus*, *stratus*, *cirro-cumulus*, *cirro-stratus*, *cumulo-stratus*, and *nimbus*. In passing, it may be of interest to note that, a few years after Howard's classification was published, an attempt was made by one Thomas Forster to introduce "popular" equivalents of these terms. Forster proposed to call cirrus "curlcloud," cumulus "stackencloud," stratus "fallcloud," etc. In other words, he assumed that because Howard's names were Latin in form they were unsuitable for use by the layman, and therefore needed to be supplemented by English names—although the proposed substitutes were, on the whole, somewhat longer and more difficult to pronounce than the originals! A parallel undertaking would be an attempt to discourage the public from calling the wind-flower "anemone," or virgin's bower "clematis." Forster's superfluous names have never taken root in our language.

The highest clouds—cirrus and cirro-stratus—are feathery in appearance, and consist of minute crystals of ice. Their altitude above sea-level averages about five miles, but is frequently much greater than this. All other clouds are composed of little drops of water—not hollow vesicles of water, as was once supposed. Neither crystals nor drops actually "float" in the air. They are constantly falling with respect to the air around them, though, as the air itself often has an upward movement, the cloud particles are not always falling with reference to the earth. In any case, their rate of fall depends upon their size, and in the case of the smaller particles is very slow. Under some conditions the particles evaporate before reaching the earth, while under others they maintain a solid or liquid form and constitute rain or snow. A fog is a cloud lying at the earth's surface.

Rainfall is one of the most important elements of climate, chiefly because of its effects upon vegetation. It is measured in terms of the depth of water that would lie on the ground if none of it ran off, soaked in, or evaporated; and this is, in practice, determined by collecting the rain, as it falls, in a suitable receiver, or rain-gauge. Usually the gauge is so shaped as to magnify the actual depth of rainfall, in order to facilitate measurement. Snow is measured in two ways; first, as snow, and, second, in terms of its "water equivalent." The latter measurement is commonly effected by melting the snow and pouring it into the rain-gauge, where it is measured as rain. By this expedient we are enabled to combine measurements of rain and snow, in order to get the total "precipitation" of a place during a given period.

Nature is notoriously partial in her distribution of this valuable element over the earth. A region having an average annual rainfall of less than ten inches is normally a desert, though irrigation or "dry-farming" methods may enable its inhabitants to practice agriculture.

The heaviest average annual rainfall in the United States (not including Alaska) is about 136 inches, in Tillamook County, Oregon. The rainiest meteorological station in the world is Cherrapunji, India, with an average of about 426 inches per annum.^[B]

[B] This is the latest official record. There are several rain-gauges at Cherrapunji, and the average amount of rain collected by any one of them varies considerably with the length of the record. Hence the widely divergent values of the rainfall at this famous station published in encyclopædias and other reference books.

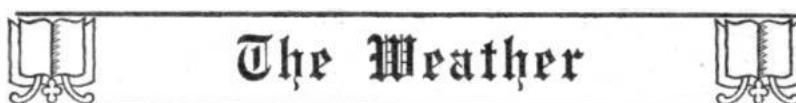
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THE OBSERVATORY ON MONTE ROSA



THE OUTPOSTS OF METEOROLOGY

Monograph Number Four in The Mentor Reading Course



HE expression used in our title seems a fitting one to apply to a number of meteorological observatories and stations maintained for the benefit of science in regions remote from the comforts and conveniences of civilization. Some are on the summits of lofty mountains, the ascent of which is laborious and even perilous. Others are situated in the bleak wildernesses of the circumpolar zones. Public attention has all too rarely been called to the heroism and self-sacrifice of the men who constitute the staffs of these lonely outposts.

The institution shown in our gravure—officially known, in honor of the Dowager Queen of Italy, as the Regina Margherita Observatory—crowns the summit of Monte Rosa, on the northern Italian frontier, and is 14,960 feet above sea-level. It is devoted not only to meteorological investigations, but to studies of the physiological effects of great altitudes and various other researches, and is open to the *savants* of all nationalities who are courageous enough to scale the second highest summit of the Alps. It is habitable for only about two months; viz., from the middle of July to the middle of September. Each year a temporary telephone line is constructed connecting the observatory with the plains of Italy. This is the highest telephone line in the world, and its installation is an arduous undertaking. A permanent line is impossible, on account of the shifting of the glaciers and snowfields on which the poles must be erected.

There is also a meteorological observatory on Mont Blanc, but it is not at the summit and is not quite so high as that on Monte Rosa. The solar observatory which once stood at the very top of Mont Blanc no longer exists. The United States Signal Service (now the Weather Bureau) formerly maintained observatories on Pike's Peak (14,134 feet) and Mount Washington (6,280 feet). The loftiest of meteorological stations was, however, that formerly operated by Harvard College Observatory on the summit of El Misti, Peru (19,200 feet).

For a number of years the United States Weather Bureau maintained a large and important observatory at Mount Weather, at the crest of the Blue Ridge, near Bluemont, Virginia. In the Old World one of the most famous of mountain meteorological observatories was that which stood on Ben Nevis (4,406), the highest summit in the British Isles. This was closed in 1904.

If the conditions of life at these high-level stations are such as to repel any but the ardent lover of science, the same is true in even greater measure of those endured by the little band of meteorologists who man the observatory maintained by the government of Argentina at Laurie Island, in the South Orkneys, on the verge of the Antarctic. Every year a party of four is sent out from Buenos Aires to spend a year of exile in this inhospitable spot, which is generally ice-bound, and has not even wireless communication with the rest of the world. This station has been in operation since 1904. The staff, which is changed each year, has embraced men of several nationalities—Scotch, American and others.

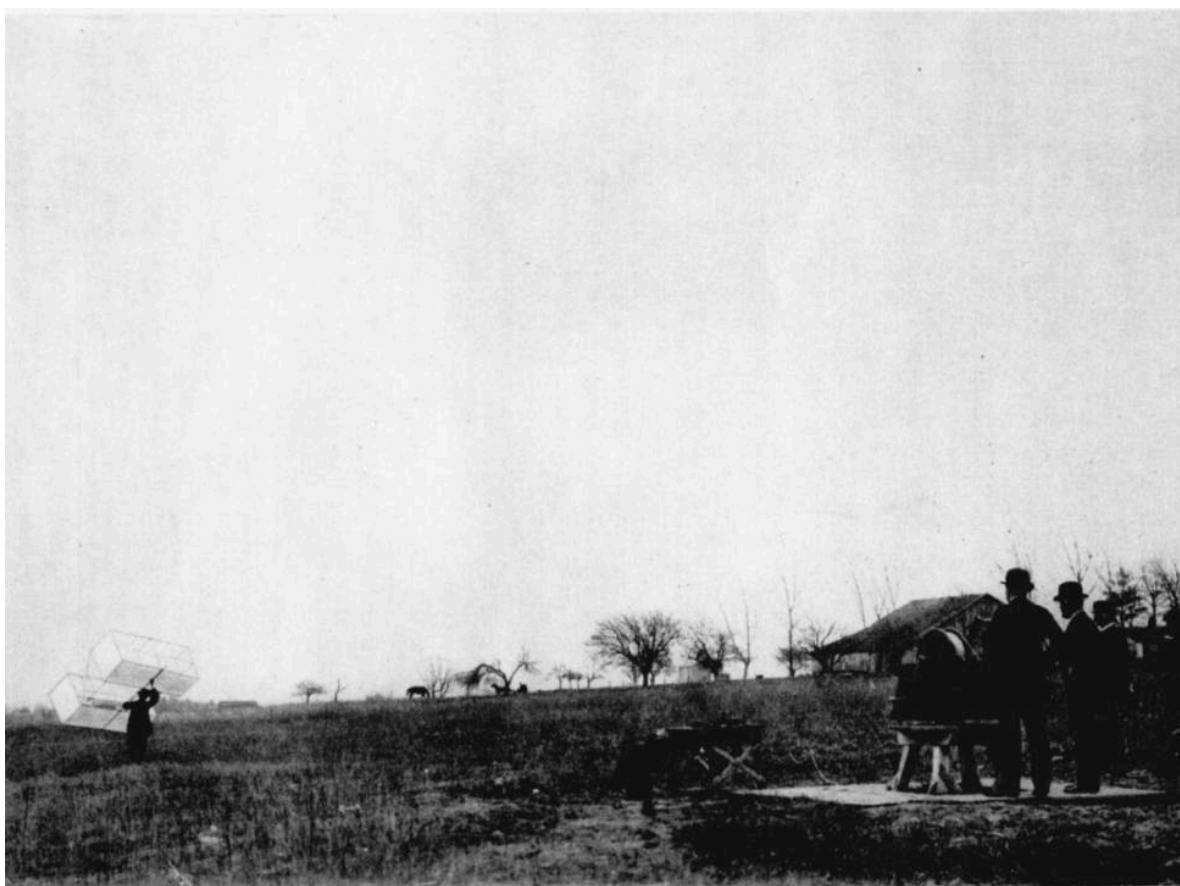
Far within the Arctic Circle two meteorological observatories are maintained in Spitsbergen; but these are, at least, connected with the world by radiotelegraphy.

If the hopes of explorer Peary are accomplished, an observatory will, one of these days, be established at the South Pole.

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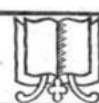
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LAUNCHING A METEOROLOGICAL KITE



The Weather



THE AIR ABOVE US

Monograph Number Five in The Mentor Reading Course



ETEOROLOGISTS are not content to limit their investigations to the stratum of air lying close to the earth's surface. Even before the demands of the aeronaut for information concerning the structure and phenomena of the atmosphere far overhead became pressing, many efforts had been made to secure such information, in view of its important bearing upon many scientific problems. As long ago as the year 1784 a balloonist, equipped with various meteorological instruments, made an ascent from London and brought back an interesting series of observations, which were communicated to the Royal Society. For more than a century the manned balloon was the principal means of sounding the upper atmosphere.

Nowadays, as a rule, the meteorologist, instead of going aloft in person, sends up a kite or a balloon to which are attached automatically registering instruments. When the aerial vehicle returns to earth its record shows in detail the conditions encountered during the journey.

Everybody remembers how Franklin brought lightning from the clouds; but it is a far cry from the simple apparatus that served Franklin's purpose to the "box kite" of modern meteorology. Science has perfected the kite almost beyond recognition. It has been shorn of that crucial feature of the schoolboy article, the tail. Even the kite "string" has become several miles of steel piano wire, wound around the drum of a power-driven winch, with elaborate apparatus for recording the force of the pull, and the angles of azimuth and altitude.

Captive balloons are sometimes used for similar investigations. When, however, it is desired to attain great altitudes the meteorologist has recourse to the so-called "sounding-balloon," which is not tethered to the earth. This is usually made of india-rubber, and when launched is inflated to less than its full capacity. As it rises to regions of diminished air pressure it gradually expands, and finally bursts at an elevation approximately determined in advance. A linen cap, serving as a parachute, or sometimes an auxiliary balloon which does not burst, serves to waft the apparatus, with its delicate self-registering instruments, gently to the ground. This commonly happens many miles—sometimes two hundred or more—from the place of ascent. Attached to the apparatus is a ticket offering the finder a reward for its return, and giving instructions as to packing and shipping. Sooner or later it usually comes back; though often months after it falls. Indeed, the large percentage of records recovered, even in sparsely settled countries, is not the least remarkable feature of this novel method of research. The instruments attached to sounding-balloons register the temperature of the air, the barometric pressure, and sometimes the humidity.

By means of the sounding-balloon the air is explored to heights of twenty miles and more! The records obtained by means of these balloons have, within the past fifteen years, completely revolutionized our ideas concerning the upper atmosphere.

Still another device employed by meteorologists is the pilot-balloon. This is also a free balloon, but carries no meteorological instruments. Its motion in the air is followed by means of a theodolite, and it serves to show the speed and direction of the wind at different levels. During the winter of 1912-13 a pilot-balloon sent up from Godhavn, Greenland, by a Danish exploring expedition reached the unprecedented altitude of more than 24 miles.

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THE EFFECTS OF SNOW AND ICE—THE CAMPUS, PRINCETON UNIVERSITY



The Weather



OUR WINTERS

Monograph Number Six in The Mentor Reading Course

In the year 1781 Thomas Jefferson wrote in his "Notes on Virginia": "A change of climate is taking place very sensibly. *** Snows are less frequent and less deep. They do not often lie below the mountains more than one, two or three days, and very rarely a week. The snows are remembered to have been formerly frequent, deep, and of long continuance. The elderly inform me that the earth used to be covered with snow about three months in every year."

Probably long before the white man came to America the patriarchs of the Indian tribes regaled the young men and maidens gathered about the campfire with reminiscences of the deep snows that prevailed in a previous generation.

In short the "old-fashioned winter" is a *perennial myth*, perpetuated by a familiar process of self-delusion! The occasional periods of abundant snow make a more lasting impression upon our minds than the long intervals in which this element was scarce or lacking. The resulting misconception is promptly dissipated when we consult the weather records, which, in some parts of the country, extend back more than a century, and prove that there has been no actual change in the climate within the period they embrace.

Of course the erroneous idea is, in some cases, due to the fact that one's childhood was spent in a part of the country in which the snowfall is normally heavier than in that where one has recently lived. The average yearly snowfall over the New England States, New York, and the borders of the Great Lakes is from 50 to 100 inches, and upward. Over the North Central States it is much less. In the Southern tier of States and along almost the whole of our Pacific coast snow is a rarity. The heaviest snowfall in this country probably occurs in the high Sierra Nevada of California, near the border of Nevada. In some places in these mountains more than 40 feet of snow falls in an average winter, while more than 65 feet has been recorded in extreme cases. Here it is a common occurrence for one-story houses to be buried, to the eaves, or above. The Southern Pacific Railway, which intersects this region, has built 32 miles of snowsheds, at a cost of \$42,000 a mile over single track and \$65,000 a mile over double track. In an average year \$150,000 is spent on these sheds in upkeep and renewals. Flat-roofed houses are unknown in this vicinity; all roofs are gabled at a sharp angle to shed the snow.

A picturesque feature of our American winters is the "ice storm," so enthusiastically described by Mark Twain:

"... When a leafless tree is clothed with ice from the bottom to the top—ice that is as bright and clear as crystal; when every bough and twig is strung with ice-beads, frozen dew-drops, and the whole tree sparkles cold and white, like the Shah of Persia's diamond plume."

Such is the artist's view of the phenomenon; but, alas! these same ice storms cause endless inconvenience and heavy expense every winter to the electrical industries, by breaking wires.

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THE WEATHER ***

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